

Operating System Concepts

UNIT I

2 Marks Questions

Define an Operating System?

An Operating System (OS) is system software that manages computer hardware, software resources, and provides common services for computer programs.

What is a process?

A process is a program in execution, including the program counter, variables, and system resources.

What are the drawbacks of Batch processing System?

- No interaction between user and job during execution
- Long wait time for job completion
- Errors detected late
- Inefficient for small jobs.

What is Time sharing system and Multiprogramming System?

- Time sharing system: Multiple users share system resources simultaneously by time-slicing CPU.
- Multiprogramming system: Multiple programs are loaded in memory and CPU switches among them to maximize utilization.

What is Real time system?

A system that responds to input instantly or within a guaranteed time frame to control physical processes.

Write the two types of Real time system.

- Hard real-time system
- Soft real-time system

Define distributed system.

A system where multiple computers communicate and cooperate to achieve a common goal appearing as a single system to users.

What is multiprocessor system?

A computer system with two or more processors that share memory and peripherals.

Give two advantages of multiprocessor system?

- Increased throughput
- Enhanced reliability

Write the two types of multiprocessor system.

- Symmetric Multiprocessing (SMP)
- Asymmetric Multiprocessing (AMP)

Define symmetric multiprocessor system.

All processors share memory and I/O equally and perform tasks independently.

Define asymmetric multiprocessor system.

Each processor is assigned a specific task and only one processor controls the system.

List any two functions of Process Management Component of the OS.

- Process creation and deletion
- Process scheduling

List any two functions of Memory Management Component of the OS.

- Allocation and deallocation of memory
- Memory protection

List any two functions of File Management Component of the OS.

- File creation and deletion
- Directory management

List any two functions of Input/Output System Management Component of the OS.

- Device driver management
- I/O scheduling

List any two functions of Networking Management Component of the OS.

- Network communication control
- Resource sharing

Write any four operating system services.

- Program execution
- I/O operations
- File system manipulation
- Communication

Give any four attributes of File.

- Name
- Type
- Location
- Size

Give any four File Types

- Text files
- Binary files
- Executable files
- Directory files

Give any four File Operation.

- Create
- Read
- Write
- Delete

Give any four Directory Operation.

- Create directory
- Delete directory

- List directory contents
- Rename directory

List any two file access methods.

- Sequential access
- Random access

Define sequential access with respect to files.
Accessing file records one after another in order.

Define random access with respect to files.
Accessing file records in any order directly.

List four types of file accesses.

- Sequential
- Direct
- Indexed
- Hashed

List any four allocation methods.

- Contiguous allocation
- Linked allocation
- Indexed allocation
- Clustered allocation

Define contiguous allocation.
Allocating consecutive blocks of storage to a file.

What is linked allocation?
File blocks are linked using pointers and may be scattered anywhere on the disk.

What is indexed allocation?
An index block contains pointers to the file's blocks, allowing direct access.

List four Free space Management.

- Bit vector
- Linked list
- Grouping
- Counting

UNIT II

Define logical and physical address.

Logical address: The address generated by the CPU during a program's execution.

Physical address: The actual address in main memory.

What is relocation register? Why it is used.

A register that holds the base address of the process in memory; used to map logical addresses to physical addresses.

What is memory management unit?

Hardware that translates logical addresses to physical addresses using relocation registers or page tables.

What is virtual memory?

A memory management technique that allows execution of processes larger than physical memory by using disk space.

Define swapping.

Moving processes between main memory and secondary storage to free up memory.

What is memory protection?

Mechanisms to prevent unauthorized access to memory by processes.

What is first-fit, best-fit, and worst-fit Memory Allocation?

- ✓ First-fit: Allocates first suitable free block.
- ✓ Best-fit: Allocates smallest free block that fits.
- ✓ Worst-fit: Allocates largest free block.

What is fragmentation? List their types.

Wastage of memory — types: internal and external fragmentation.

What is paging and Page Fault?

Paging: Dividing memory into fixed-size pages to allow non-contiguous allocation.

Page fault: When a requested page is not in memory.

What is segmentation?

Dividing a program into variable-sized segments based on logical divisions.

What is segmentation hardware?

Hardware that maps logical addresses (segment number and offset) to physical addresses.

What is demand paging?

Loading pages into memory only when they are needed.

What is pure demand paging?

No pages are loaded initially; all pages are loaded on demand.

List two hardware to support demand paging.

- ✓ Page table
- ✓ Translation Lookaside Buffer (TLB)

What are the major problems with implementing Demand Paging?

- ✓ Page fault overhead
- ✓ Thrashing

Define thrashing and virtual memory.

Thrashing: Excessive paging causing low CPU utilization.

Virtual memory: Memory abstraction extending RAM using disk.

Differentiate between logical and physical address space.

Logical address space is generated by the CPU; physical address space is actual memory.

Differentiate between the paging and segmentation.

Paging uses fixed-size pages; segmentation uses variable size segments.

What are the limitations of swapping?

High overhead and slow if done excessively.

Differentiate swapping and demand paging.

Swapping moves entire process; demand paging loads pages as needed.

What is the basic approach of Page Replacement?

Replacing pages in memory when new pages need to be loaded.

What is FIFO page replacement algorithm?

Removes oldest loaded page first.

What is Optimal Page replacement algorithm?

Replaces page that won't be used for the longest time in future.

What is LRU Page replacement algorithm?

Replaces the page least recently used.

What do you mean by Belady's anomaly?

Increasing number of frames causes more page faults in FIFO.

Define Working-Set Model.

Set of pages a process is currently using.

What is Page Fault Frequency?

Rate at which page faults occur.

What is Disk Scheduling?

Ordering disk I/O requests to reduce seek time.

What is FCFS scheduling?

Processes requests in order received.

What is SSTF scheduling?

Selects request closest to current head position.

What is SCAN scheduling?

Moves head in one direction servicing requests until end, then reverses.

What is C-SCAN scheduling?

Like SCAN but jumps back to beginning after reaching end.

What is LOOK scheduling?

Like SCAN but reverses direction at last request, not disk end.

UNIT III

2 Marks Questions

What is a process? List the four components of the process.

A process is a program in execution.

Components:

- ✓ Program code
- ✓ Program counter
- ✓ CPU registers
- ✓ Process stack

Differentiate process and program.

A program is a passive set of instructions; a process is an active execution of a program.

List out any four process states.

- ✓ New
- ✓ Ready
- ✓ Running
- ✓ Waiting

What is PCB? List its component.

Process Control Block stores process information.

Components:

- Process ID
- Program counter
- CPU registers
- Process state
- Memory limits
- Accounting information

What is Context switch?

Saving state of a running process and loading state of another process.

Define Thread.

Smallest unit of execution within a process.

Define dispatcher.

Module that gives control of CPU to process selected by scheduler.

Define independent and cooperating processes.

Independent: Cannot affect or be affected by others.

Cooperating: Can affect or be affected by others.

What do you mean by Inter Process communication (IPC)?

Mechanism allowing processes to communicate and synchronize.

What is a scheduler? Mention their types.

Software that selects process for execution.

Types:

- Long-term scheduler
- Short-term scheduler
- Medium-term scheduler

What is ready queue and device queue?

Ready queue: Processes ready for CPU.

Device queue: Processes waiting for I/O devices.

Define CPU and I/O Burst.

CPU burst: Time CPU is actively used.

I/O burst: Time process waits for I/O.

Define throughput and turnaround time.

Throughput: Number of processes completed per unit time.

Turnaround time: Time from submission to completion.

Define response time and arrival time.

Response time: Time from submission to first response.

Arrival time: Time process enters ready queue.

What is waiting time?

Total time a process spends in ready queue.

Define preemptive scheduling and non-preemptive scheduling.

- Preemptive: Process can be interrupted.
- Non-preemptive: Process runs till completion or waiting.

Difference between job and process.

Job is a program submitted for execution; process is an executing instance.

What is aging?

Technique to prevent starvation by increasing priority over time.

List the disadvantages of SJF.

Difficulty in knowing next CPU burst

Possible starvation of long processes

Differentiate long term schedulers and short terms scheduler.

Long-term: Controls admission of jobs.

Short-term: Controls CPU allocation.

UNIT IV

2 Marks Questions

Define critical section problem.

Problem of ensuring that multiple processes do not simultaneously execute critical section accessing shared resources.

What is the Requirement of critical section problem?

- Mutual exclusion
- Progress
- Bounded waiting

Write General structure of a typical critical section problem.

Entry section → Critical section → Exit section → Remainder section

What is semaphore?

Synchronization tool using integer value for process signaling.

Write the General structure of wait() and Signal().

wait(): Decrements semaphore; if negative, process blocks.

signal(): Increments semaphore; if zero or negative, wakes a waiting process.

What is deadlock? Give the real-life example for deadlock.

Situation where processes wait indefinitely for resources held by each other. Example:

Two cars blocking a narrow bridge from opposite directions.

Which are the Necessary Conditions for Deadlock?

- Mutual exclusion
- Hold and wait
- No preemption
- Circular wait

What is Resource-Allocation Graph?

Graph showing allocation and request of resources by processes.

Mention the methods for handling deadlocks.

- Deadlock prevention
- Deadlock avoidance
- Deadlock detection and recovery

Explain the term safe state.

A state where system can allocate resources to each process in some order without deadlock.

What is race condition?

Situation where behavior depends on sequence or timing of other uncontrollable events.

What is deadlock prevention?

Structuring system to negate one of the necessary conditions for deadlock.

Mention two ways of recovery from deadlock.

- Process termination
- Resource preemption

Mention different ways to eliminate resource preemption.

- Rollback
- Process resource release

Mention different ways to eliminate deadlock by process termination.

- Terminate all deadlocked processes
- Terminate processes one by one until deadlock breaks